Python Functions

```
def fun():
    print("Welcome to GFG")
```

Calling a Function

```
def fun():
    print("Welcome to GFG")

fun()
```

```
Welcome to GFG
```

Function Arguments

```
def evenOdd(x):
    if (x % 2 == 0):
        return "Even"
    else:
        return "Odd"

print(evenOdd(16))
print(evenOdd(7))
```

```
Even
Odd
```

Default argument

```
def myFun(x, y=50):
    print("x: ", x)
    print("y: ", y)

myFun(10)
```

```
x:  10
y:  50
```

Keyword Arguments

```
def student(fname, lname):
    print(fname, lname)

student(fname='Geeks', lname='Practice')
student(lname='Practice', fname='Geeks')
```

```
Geeks Practice
Geeks Practice
```

Positional Arguments

```
def nameAge(name, age):
    print("Hi, I am", name)
    print("My age is ", age)

print("Case-1:")
nameAge("Olivia", 27)

print("Case-2:")
nameAge(27, "Olivia")
```

```
Case-1:
Hi, I am Olivia
My age is  27
Case-2:
Hi, I am 27
My age is  Olivia
```

Arbitrary Arguments

```
def myFun(*args, **kwargs):
    print("Non-Keyword Arguments (*args):")
    for arg in args:
        print(arg)
```

```
    print("Keyword Arguments (**kwargs):")
    for key, value in kwargs.items():
        print(f"{key} == {value}")

myFun('Hey', 'Welcome', first='Geeks', mid='for', last='Geeks')
```

```
Non-Keyword Arguments (*args):
Hey
Welcome
Keyword Arguments (**kwargs):
first == Geeks
mid == for
last == Geeks
```

Function within Functions

```
def f1():
    s = 'I love GeeksforGeeks'
    def f2():
        print(s)

    f2()
f1()
```

```
I love GeeksforGeeks
```

Return Statement

```
def sq_value(num):
    return num**2

print(sq_value(2))
print(sq_value(-4))
```

```
4
16
```

Pass by Reference and Pass by Value

```
def myFun(x):
    x[0] = 20
```

```
    x[0] = 20

b = [10, 11, 12, 13]
myFun(b)
print(b)

def myFun2(x):
    x = 20

a = 10
myFun2(a)
print(a)
```

```
[20, 11, 12, 13]
10
```